Peak coordinates and spatial maps

The 29 ICs were categorized into these three distinct functional networks based on anatomical and other prior knowledge: the default mode network (DMN; 7 ICs), the cognitive control network (CC; 17 ICs), and the subcortical network (SC; 5 ICs). The spatial distribution of the 29 ICs within the functional networks is illustrated in Figure 1. Using Group ICA/IVA of the fMRI Toolbox GIFT v4.0.6.0, and guided by prior knowledge in the field, these 29 ICs were mapped to brain regions. Functional network regions were assigned according to the MNI coordinate system (x, y, z). Table 1 provides detailed information on the functional networks, peak coordinates, and spatial maps for the 29 ICs. The interactions among the ROIs with functional networks SN, CEN, and DMN are highlighted. ICs 27, 29, 39, and 40 (from CC) and ICs 45 and 47 (from DMN) belong to the SN, while ICs 26, 29, 31–36, 38, 40, and 41 also contribute to the CEN within the CC domain.

Table 1: Detailed information for the 29 ICs selected from NeuroMark among 53 intrinsic networks, showing corrected ROIs with network interactions. Color-coded as follows: CEN = blue, SN = red, DMN = forest, mixed = purple.

Selected Components as Regions of Interest	**X**	**Y**	**Z**	**Interaction**
Subcortical network (SC)				
Caudate (1)	6.5	10.5	5.5	CEN/DMN
Subthalamus/Hypothalamus (2)	-2.5	-13.5	-1.5	SN
Putamen (3)	-26.5	1.5	-0.5	CEN
Caudate (4)	21.5	10.5	-3.5	CEN
Thalamus (5)	-12.5	-18.5	11.5	SN/CEN
Cognitive Control Network (CC)				
Inferior Parietal Lobule (lIPL, 26)	45.5	-61.5	43.5	CEN
Insula (27)	-30.5	22.5	-3.5	SN
Superior Medial Frontal Gyrus (SMFG, 28)	-0.5	50.5	29.5	CEN
Inferior Frontal Gyrus (IFG, 29)	-48.5	34.5	-0.5	CEN/SN
Right Inferior Frontal Gyrus (R IFG, 30)	53.5	22.5	13.5	CEN
Middle Frontal Gyrus (MiFG, 31)	-41.5	19.5	26.5	CEN
Inferior Parietal Lobule (lIPL, 32)	-53.5	-49.5	43.5	CEN
Right Inferior Parietal Lobule (R IPL, 33)	44.5	-34.5	46.5	CEN
Supplementary Motor Area (SMA, 34)	-6.5	13.5	64.5	CEN
Superior Frontal Gyrus (SFG, 35)	-24.5	26.5	49.5	CEN
Middle Frontal Gyrus (MiFG, 36)	30.5	41.5	28.5	CEN
Hippocampus (HiPP, 37)	23.5	-9.5	-16.5	DMN
Left Inferior Parietal Lobule (L IPL, 38)	-47.5	5.5	22.5	CEN/DMN
Middle Cingulate Cortex (MCC, 39)	-15.5	20.5	37.5	SN
Inferior Frontal Gyrus (IFG, 40)	39.5	44.5	-0.5	CEN/SN
Middle Frontal Gyrus (MiFG, 41)	-26.5	47.5	5.5	CEN
Hippocampus (HiPP, 42)	-24.5	-36.5	1.5	DMN
Default Mode Network (DMN)				
Precuneus (43)	-8.5	-66.5	35.5	DMN
Precuneus (44)	-12.5	-54.5	14.5	DMN
Anterior Cingulate Cortex (ACC, 45)	-2.5	35.5	2.5	SN/DMN
Posterior Cingulate Cortex (PCC, 46)	-5.5	-28.5	26.5	DMN
Anterior Cingulate Cortex (ACC, 47)	-9.5	46.5	-10.5	SN/DMN
Precuneus (48)	-0.5	-48.5	49.5	DMN
Posterior Cingulate Cortex (PCC, 49)	-2.5	54.5	31.5	DMN

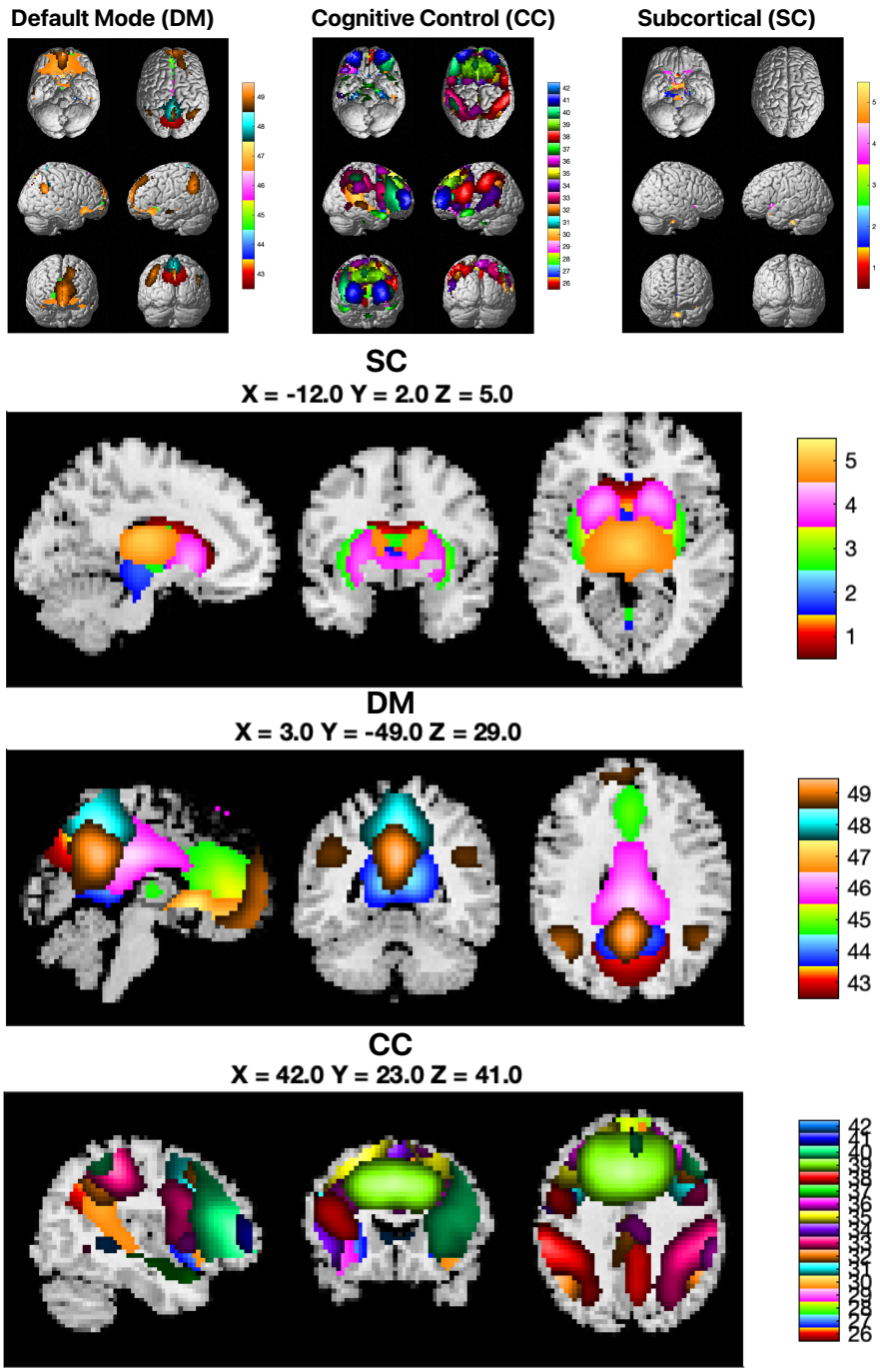

Figure 1: Spatial maps of the 29 selected ICs, including 5 subcortical components (1–5), 7 default mode components (43–49), and 17 cognitive control network components (26–42).